

Sensing with Spikes: Neuromorphic Optical Tactile Sensor for Contact Force and Position Detection

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Abstract—We present a neuromorphic optical tactile sensor capable of detecting contact force and position across a 40 mm sensing area using only 7 channels of event-driven digital spike signals. Unlike prior neuromorphic tactile sensors that rely on exotic materials such as memristors or PiezoFETs, or bulky event-based cameras, our design is built entirely from off-the-shelf components — IR LEDs and phototransistors embedded in silicone rubber — eliminating the need for ADC conversion or microcontroller-based polling. The sensor operates on an analog Leaky Integrate-and-Fire (LIF) circuit that generates sparse voltage spikes directly from optical signals, where spike firing rate encodes applied force and spatial firing patterns encode contact location. To decode these signals, we employ a Spiking Neural Network (SNN) trained on 714 recordings across multiple probe sizes. The best configuration achieves a force estimation MAE of 0.73 N and a mean distance error of 3.3 mm, demonstrating that rich tactile information can be extracted from a sparse, low-channel neuromorphic front-end.

I. INTRODUCTION

Can robots possess tens of thousands of tactile sensors distributed across their bodies like humans? While many robotic tactile sensors focus on fingertip sensing for manipulation tasks [1], contact-rich and unstructured environments necessitate understanding tactile information from other parts of the robotic body, such as the dorsal surface of the hand [2], [3], arms [4]–[6], and torsos [7], [8] to detect contact with objects or the environment. Social robots and those used in healthcare, which interact with humans through touch, require haptic communication as a vital form of interaction [6]. Moreover, when exerting large forces, such as holding a patient [9], it becomes critical to understand contact information from the arms and torso to ensure safety.

Equipping tactile sensors on different parts of the robot's body presents processing challenges. Managing tactile sensors over larger contact areas can be challenging. It may require distributed microcontrollers [6]–[8] to continuously monitor tactile information, necessitating persistent communication between each local processor and the main robot computing unit, which can lead to significant communication overhead.

Unlike these robotic systems, human mechanoreceptors generate voltage spikes only when contact is made. These sensors activate autonomously, sending signals to the central nervous system without the need for continuous communication. In this regard, neuromorphic tactile sensors represent a promising approach to building a human-like tactile sensor network. Mimicking this, neuromorphic robotic

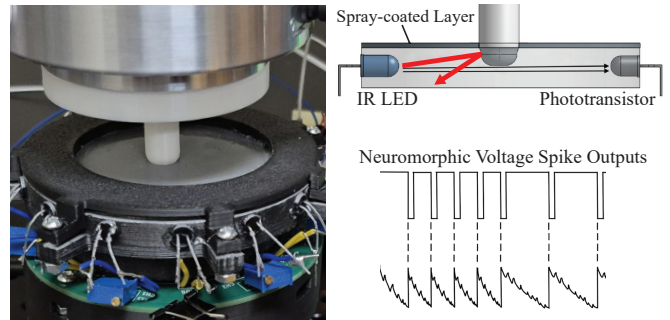


Fig. 1: Neuromorphic optical tactile sensor. (Top right) When the sensor is indented by a probe, the spray-coated surface of the clear silicone rubber deforms, altering the light intensity transmitted to the phototransistor. (Bottom right) The phototransistor output is fed into an artificial neuron circuit, generating voltage spikes.

tactile sensors have been studied where event driven signals generated using lab-developed materials like memristors [10] and PiezoFET [11] that may not be widely accessible, or event-based cameras [12], [13] which are somewhat bulky to be use for scalable tactile skin.

In this paper, we introduce a neuromorphic tactile sensor built entirely from off-the-shelf optical components (IR LEDs and phototransistors), covering a 40 mm diameter sensing area while outputting 7 channels of event-driven digital spike signals. As shown in Fig. 1, the system employs an optical transduction mechanism in which a Leaky Integrate-and-Fire (LIF) analog circuit generates digital voltage spikes directly, without requiring ADC conversion or continuous microcontroller polling. The spike firing rate increases with applied pressure, and the spatial firing pattern varies across different contact locations. Using these sparse spike signals, a spiking neural network (SNN) is trained to achieve a force estimation mean absolute error (MAE) of 0.73 N over a 15 N range and a contact position mean distance error (MDE) of 3.3 mm.

A. Related Work

Event-driven tactile sensing has advanced along two complementary directions: spike-native tactile transduction at the hardware level, and asynchronous communication architectures for scalable tactile skins.

Spike-native tactile transduction focuses on converting mechanical stimuli directly into spike trains at the circuit or device level. PiezoFET-based taxels integrated with leaky integrate-and-fire (LIF) circuits demonstrate force-dependent spike-rate modulation and adaptation dynamics [11], [14]. More recently, diffusive memristors have been exploited to achieve threshold-triggered spiking directly at the device

level [10]. Despite these advances, most such systems depend on custom or exotic materials and are validated only on single taxels, leaving scalability to practical sensing areas as an open challenge.

A parallel line of work targets scalable, low-latency communication for tactile skins. Architectures such as ACES [15] and event-driven tactile hubs [16] employ asynchronous address-event representation to achieve sparse, bandwidth-efficient data transfer across large arrays. However, in these systems tactile signals are typically digitized via ADC sampling before event detection; the communication is sparse, but the transduction itself remains sampling-based, meaning the ADC overhead scales with array size and sampling persists even during idle conditions — undermining the efficiency gains of sparse communication.

On the optical side, event-camera-based tactile sensors such as NeuroTac and Evetac [12], [13] generate asynchronous visual event streams that capture fine surface deformations. While powerful, these approaches depend on imaging hardware and high-dimensional visual data pipelines, limiting deployability in compact or resource-constrained systems. Fiber Bragg Grating (FBG)-based optical tactile sensing has also been combined with spiking neural networks to achieve high-accuracy contact localization on large-area e-skins [17], [18]. These systems demonstrate compelling neuromorphic processing, but spike conversion occurs in software or on a downstream neuromorphic processor after analog-to-digital interrogation of the optical signals — the transduction front-end itself remains sampling-based.

To our knowledge, this work presents the first tactile sensor front-end that generates neuromorphic spike signals directly at the analog circuit level using purely optical transduction, without ADC conversion. Rather than relying on custom materials, event cameras, or optical interrogation equipment, our design is built from off-the-shelf components — LEDs and phototransistors — that natively produce sparse spike trains from contact events. Complementing this front-end, we introduce a sparse neuromorphic tactile sensing system that jointly infers contact force and location from spike timing and rate across only 7 channels, without requiring array-based spatial coverage. Together, these contributions offer a practical and deployable sensing solution for neuromorphic robotics.

II. DESIGN

A. Neuron-Inspired Circuit Model

In the biological nervous system, action potentials, which are digital-like signals, are transmitted through neurons. One widely used model to explain the generation of action potentials is the Leaky Integrate-and-Fire (LIF) neuron model [19], as shown in Fig. 2a. In this model, the neuron's membrane is represented by a capacitor and a leaky resistor, while the flow of ionic charge is modeled as a current source. As input charges accumulate across the membrane (i.e., integrated by the capacitor), the membrane potential increases. When this potential reaches a certain threshold, the neuron fires an

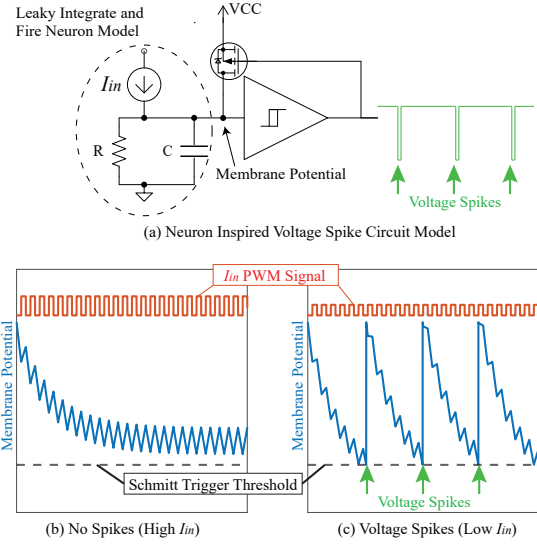


Fig. 2: Artificial neuron circuit model. (a) The neuron's membrane potential is connected to a Schmitt trigger, resetting via a P-MOSFET when a voltage spike occurs. (b) When the input current is high, the membrane potential stays above a 1.1 V threshold. (c) When low, the potential drops below the threshold, triggering voltage spikes.

action potential. After firing, the membrane potential is reset to a lower value, often referred to as the reset potential.

Inspired by this model, we designed an artificial neuron circuit to generate voltage spikes. Although such embodiments can be realized with advanced materials like memristors [10], we opted for a more accessible architecture leveraging standard, commercially available components. As shown in Fig. 2a, our design simulates the LIF neuron model, connected to a Schmitt trigger [20]. The lower threshold of the Schmitt trigger can be easily adjusted by tuning a variable resistor. Unlike the biological model, our circuit generates voltage spikes in the negative direction: the line is held high at V_{CC} , and each spike momentarily drives the signal low (ground). These spikes turn on a P-MOSFET, which resets the membrane potential back to V_{CC} . This configuration is designed to interface with the mechanical setup that will be discussed in Section II-B.

In this design, we pulse the current input (I_{in}) using Pulse Width Modulation (PWM). The amplitude of I_{in} decreases when an external force is applied. When there is no contact (i.e., high I_{in}), the charging (on-phase) and discharging (off-phase) of the capacitor balance, maintaining a steady-state voltage above the Schmitt trigger threshold, preventing voltage spikes (Fig. 2b). When contact force is applied (i.e., low I_{in}), the charging phase cannot counterbalance the discharge, causing the voltage to drop below the threshold, triggering the comparator output to go low, which in turn resets the membrane potential to V_{CC} (Fig. 2c). The level of I_{in} sets the interval between spikes: a high I_{in} results in longer intervals (lower spike rate), while a low I_{in} results in shorter intervals (higher spike rate), as shown in Fig. 3.

The steady-state value of the potential can be computed using a simple RC circuit model. When I_{in} is in the on-phase and the initial potential is V_0 , the potential after time

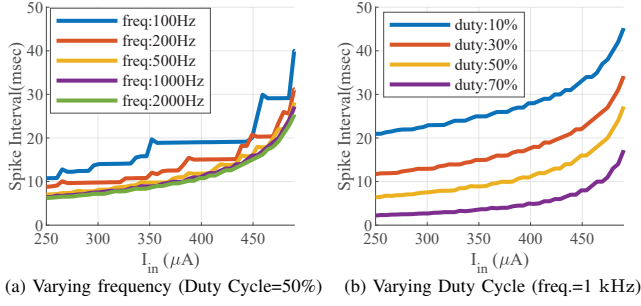


Fig. 3: Simulation of voltage spike intervals from the artificial neuron, depending on I_{in} 's (a) PWM frequency and (b) duty cycle. $R = 7.5k\Omega$ and $C = 1\mu F$.

t_{on} is:

$$V_1 = V_0 \cdot e^{-\frac{1}{RC}t_{on}} + R \cdot I_{in} \cdot \left(1 - e^{-\frac{1}{RC}t_{on}}\right) \quad (1)$$

During the off-phase (t_{off}), the potential becomes:

$$V_2 = V_1 \cdot e^{-\frac{1}{RC}t_{off}} \quad (2)$$

At steady state, $V_2 = V_0$, allowing us to compute V_0 , the steady-state voltage. Given the value of I_{in} without contact, the Schmitt trigger threshold can be set just below this steady-state voltage, V_{0ss} .

Similarly, for a given threshold value, we can estimate when the voltage will cross the threshold by sequentially computing the voltage progression (Eqs. (1) and (2)), as shown in Fig. 2c. Using this model, we can simulate the effect of I_{in} 's PWM frequency and duty cycle. As shown in Fig. 3a, when the PWM frequency is low (e.g., ≤ 200 Hz), the I_{in} vs. spike interval relationship exhibits a step-wise graph. This implies that different I_{in} values (or input stimuli) cannot be discerned based on the voltage spike pattern alone. We believe this step-wise behavior arises from the fact that a long off-time compared to the RC time constant makes the system predominantly affected by the discharge phase, rather than the charging phase, where I_{in} plays a role. The frequency of I_{in} should be high enough relative to the RC time constant to ensure a more monotonic relationship between I_{in} and the spike interval. With a sufficient PWM frequency (e.g., 1 kHz as in Fig. 3b), a lower duty cycle (shorter on-time) produces a longer spike interval, with greater sensitivity ($\Delta t_{spike\ interval}/\Delta I_{in}$). This occurs because a lower duty cycle reduces the threshold level, leading to a greater voltage drop from V_{CC} , which requires more charge-discharge cycles to return to the threshold, thus making the system more sensitive to changes in I_{in} . However, a lower duty cycle tends to result in longer intervals, which may not update changes in inputs quickly enough.

B. Mechanical Design and Fabrication

To implement the artificial neuron model as a tactile sensor, we designed an optical sensor that reduces input current in response to external forces, inspired by previous work [21]. This design utilizes low-cost components, including IR-LEDs, phototransistors, and silicone rubber. As illustrated in Fig. 1, the LEDs and phototransistors are embedded in transparent silicone rubber, with the surface covered by a

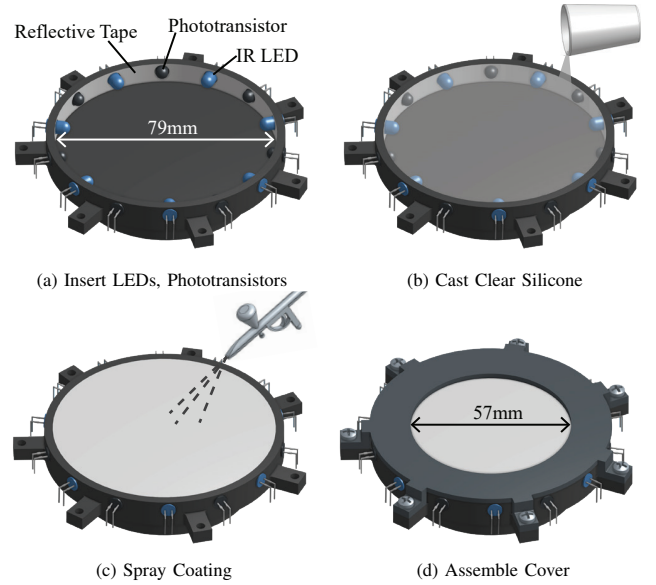


Fig. 4: Sensor fabrication process: (a) IR LEDs and phototransistors inserted into a 79 mm 3D printed circular housing with reflective Mylar lining, (b) clear silicone rubber cast and cured, (c) Spray coating applied via airbrush, and (d) protective cover assembled, exposing a 57 mm active contact area. Silicone thickness: 8 mm; LED/phototransistor depth from top surface: 3.3 mm.

reflective layer. When the surface deforms under contact force, part of the light is reflected, reducing the intensity of light reaching the phototransistor, thereby decreasing its output current. The LED input is modulated via PWM to generate a corresponding PWM output current.

As shown in Fig. 4, the sensor fabrication process involves: (a) Seven pairs of IR LEDs (WP7113SF6BT-P2) and phototransistors (SFH309FA) are evenly distributed around a 3D-printed circular housing in an alternating arrangement. Reflective metalized polyester Mylar film is attached along the inner wall, allowing each LED to influence not only its paired phototransistor through direct illumination but also neighboring phototransistors through reflected light, enriching the spatial response of the sensor. (b) Clear silicone rubber (XP-565, Silicones Inc.) is then poured using a non-standard 100:6 mixing ratio for softer material, degassed for 1 hour, and cured for 24 hours. (c) A coating layer is applied by airbrushing a mixture of silicone base (Smooth-On, PsychoPaint), metallic ink, and silicone solvent (Smooth-On, NOVOCS), followed by an additional 24-hour cure. (d) Finally, a cover is assembled to prevent excessive contact at the periphery, which could otherwise cause silicone delamination.

C. Electrical Design

We integrate the LEDs and phototransistors into the artificial neuron model, as shown in Fig. 5. A Schmitt trigger is implemented using a comparator (TLV3201AIDCKT), with the threshold level adjustable via the potentiometer (R3); increasing R3 reduces the threshold. The output of the comparator, which is what we measure as spike voltage, is connected to P-MOSFET (BSS84W-7-F). We design the

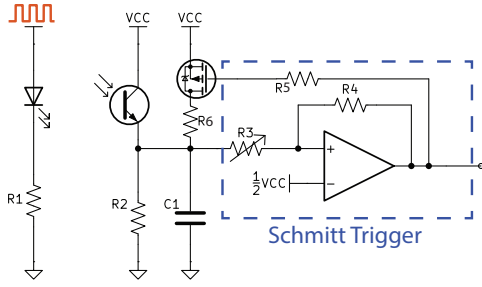


Fig. 5: Circuit schematic of a pair consisting of an IR-LED, phototransistor, and artificial neuron. $R_1=5\Omega$, $R_2=7.5k\Omega$, and $C_1=1\mu F$. R_3 and R_4 are installed on the Schmitt trigger. $R_3=50k\Omega$ potentiometer, $R_4=82k\Omega$, $R_5=R_6=1.5k\Omega$.

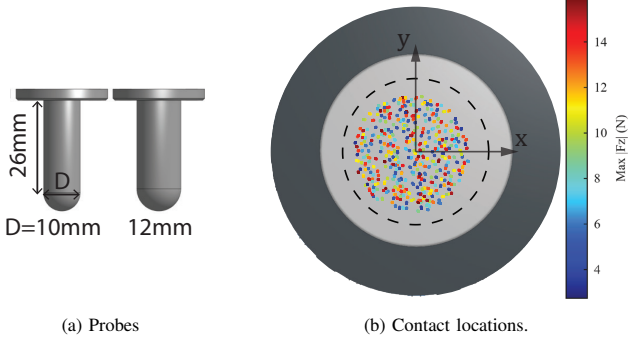


Fig. 6: (a) Two probes with cylindrical bodies and dome-shaped end-tips of varying diameters. (b) Tested probe contact locations shown as colored dots, where each dot represents the probe center and color encodes the randomly assigned press force. The dashed circle indicates the active contact region, accounting for the 12 mm probe diameter.

circuit for 7 LED-phototransistor pairs on a single PCB, including a microcontroller (CY8C4147AZI-S455). The microcontroller drives the LEDs and records the digital voltage spikes. The LEDs are driven at 2 kHz, with 7 LEDs turned on sequentially, resulting in a 14.3% duty cycle. We use input capture to record the spike timing for each channel, with a timestamp resolution of 1 μs .

III. EXPERIMENT

To characterize the sensor's response to applied forces across different locations, we conducted a series of pressing tests covering both the center and peripheral contact locations, aiming to examine how the sensor's spike outputs encode contact location and force magnitude.

A. Setup

As illustrated in Fig. 1, a robot arm (Franka Emika, Research 3) equipped with a custom 3D-printed probe was used to apply vertical forces on the sensor. Two dome-end diameters (10 and 12 mm) were tested to evaluate the sensor's response to varying contact geometries (Fig. 6a). In each trial, a single probe and target location were used, with a load cell (ATI Gamma) mounted beneath the sensor to measure the applied force. Cartesian impedance control maintained a vertical pose and ensured consistent force application over a set duration.

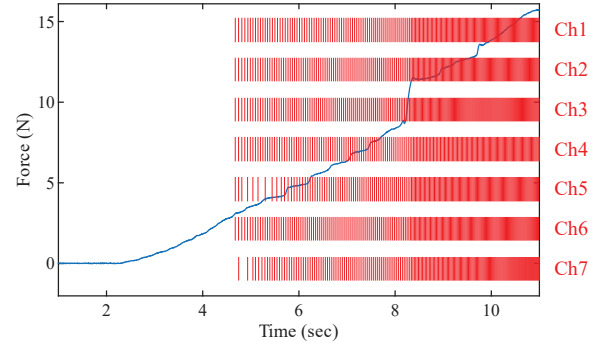


Fig. 7: Voltage spikes (red lines) from each channel while pressing a probe ($D=10$ mm) at the center of the sensor.

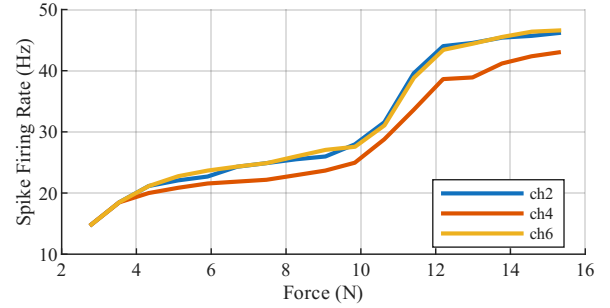


Fig. 8: Spike firing rate progression with increasing center pressing force for selected channels.

B. Center Press Test

In the center press test, a normal force of up to 15 N was applied by gradually pressing the robot arm over 10 seconds. This test was performed using the 10 mm probe.

C. Data Collection for Force and Location Estimation

To build a comprehensive dataset for contact location and force estimation, we applied 357 randomly distributed contact locations with randomly varied pressing forces for each probe, as shown in Fig. 6b. Contact locations were confined within a 42 mm diameter active region, accounting for the 12 mm probe tip within a 30 mm center boundary. For each press, the normal force was ramped up gradually over 1.5 s, held for 1.5 s, then released. The contact location in robot coordinates and the applied force measured by the load cell were recorded as time-series data throughout each pressing sequence.

IV. RESULTS

A. Sensor Outputs on Center Pressing

The center pressing results shown in Fig. 7 demonstrate that the applied force is encoded in the voltage spikes across all channels. Below 3 N, no spikes are observed, validating the sensor's event-driven behavior where spikes are produced only under meaningful load. This firing threshold can be tuned via the Schmitt trigger threshold (see Fig. 2) using the R_3 potentiometer. The minimum force threshold depends on the ambient noise level, which determines how closely the Schmitt trigger threshold can approach the steady-state non-contact value, as well as mechanical factors such as silicone softness and the LED-phototransistor-to-surface offset.

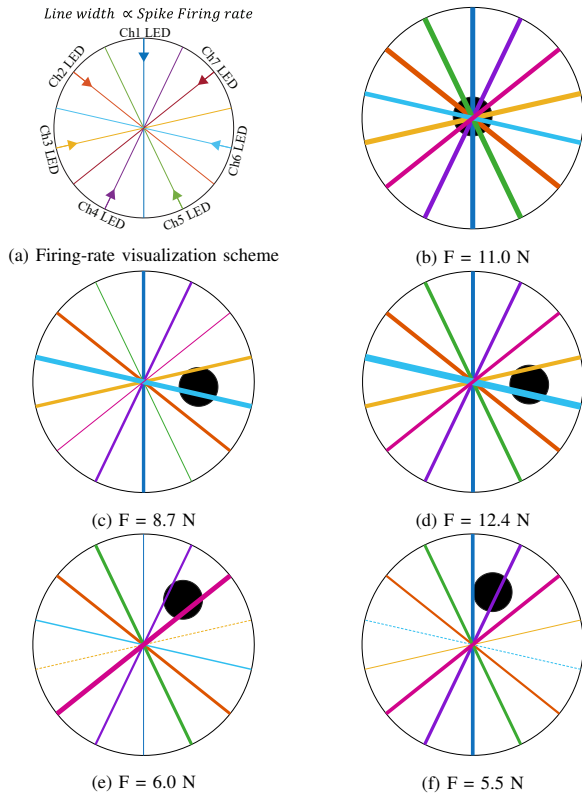


Fig. 9: Spike firing rates of each channel. The plot shows each channel's firing rate as the width of the channel line shown in (a). (b)–(f) The dotted line represents no voltage spikes, and the black circle indicates the probe's ($D=10$ mm) contact location. The contact force is noted in the caption of each plot. The line widths are proportional to the firing rate, with a consistent scaling factor across (b)–(f).

The force–spike firing rate relationship shown in Fig. 8 exhibits three distinct regions: a gradual increase in the low-force range (3–10 N), a rapid rise between 10–12 N, and saturation above ~ 14 N. This behavior is consistent with findings in [21], where initial surface deflection produces minimal light modulation, followed by a highly sensitive regime of active light blocking, and ultimately saturation as no further occlusion occurs. These characteristics offer practical guidance for sensor design: for applications requiring sensitivity in the lower force range, softer materials and a reduced LED–phototransistor-to-surface offset would be beneficial.

B. Sensor Outputs on Different Contact Locations

The spike firing patterns shown in Fig. 9 vary distinctly with contact location and force. At center contact (Fig. 9b), all channels fire at uniformly high rates, as the probe deformation disrupts all seven light paths simultaneously. For off-center contacts (Fig. 9c–f), channels whose light paths pass nearest to the contact point exhibit the strongest firing, while channels orthogonal to the contact are also activated, likely due to indirect light modulation via the reflective film walls.

As force increases at the same location (Fig. 9c to d), firing rates intensify and spread to adjacent channels. When comparable forces are applied at slightly different locations (Fig. 9e to f), the firing patterns across channels differ

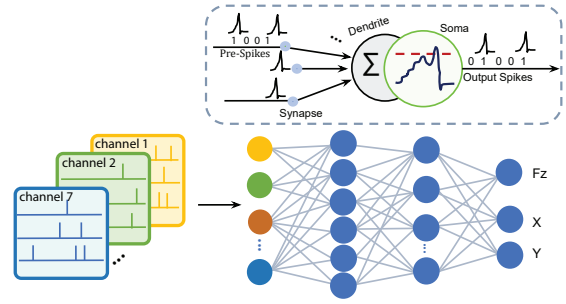


Fig. 10: Spiking neural network (SNN) architecture for contact force and location regression. Each neuron implements a bioinspired leaky integrate-and-fire (LIF) model, mapping multi-channel spike inputs to continuous estimates of F_z , X , and Y .

noticeably. Together, these results demonstrate that both the intensity and spatial pattern of spike firing are sensitive to contact location and force magnitude, providing a rich dataset for data-driven prediction. Given the nonlinear deformation of silicone rubber and the complexity of multi-path light reflection, this dataset is well-suited for learning-based estimation models.

C. Predicting Contact Location and Force

1) *Spiking Neural Network Design*: To predict contact force and location from spike signals, we employ a Spiking Neural Network (SNN), which is inherently well-suited for neuromorphic spike inputs, as both the data and the network neurons communicate via spikes [22]. We formulate the tactile decoding task as a multi-output regression problem using a fully connected SNN with Leaky Integrate-and-Fire (LIF) neurons. Input spikes from the 7-channel sensor are binned at 30 Hz (33 ms windows), with the spike count per bin fed as the network input at each timestep. Each hidden layer applies a linear projection, batch normalization, LIF spike generation, and dropout (rate = 0.1). The output is a linear layer applied to the spike activations of the final hidden layer, producing three continuous predictions: contact force F_z (N) and contact position coordinates (x, y) (mm).

For each press recording, the spike sequence is segmented using a sliding window of length t_{win} with a fixed stride of 200 ms, yielding multiple training samples per recording that span both the force ramping and plateau phases. The network is trained jointly on all three outputs by minimizing the sum of normalized mean squared errors (MSE) of F_z and (x, y) , each normalized by the training set mean and standard deviation. Performance is reported as mean absolute error (MAE, in N) for force and mean distance error (MDE, in mm), defined as $MDE = \frac{1}{N} \sum \sqrt{(\hat{x} - x)^2 + (\hat{y} - y)^2}$.

Training uses the AdamW optimizer ($\text{lr} = 1 \times 10^{-3}$, weight decay = 5×10^{-5}) for up to 300 epochs with a batch size of 512. The learning rate is halved when the validation R^2 plateaus for 10 consecutive epochs (minimum $\text{lr} = 1 \times 10^{-6}$), and early stopping is applied after 60 epochs without improvement. The dataset comprises approximately 714 recordings across two probe sizes (D10, D12), split at the recording file level (80/20 train/test) to prevent data leakage between sets. We used the snnTorch library for training [23].

TABLE I: Ablation study on network configuration, training segment window (t_{win}), and LIF decay parameter β . Results are reported on the test dataset. Each entry shows the R^2 value for force, the variance-pooled R^2 for position (jointly over x and y), the force mean absolute error (MAE), and the positional mean distance error (MDE).

Hidden Layer	$t_{win}(s)$	β	F_z (N)		X, Y (mm)	
			R^2	MAE	R^2	MDE
(256,512,256)	0.5	0.95	0.92	0.73	0.82	3.3
	0.5	0.75	0.88	0.88	0.74	4.2
	1.0	0.95	0.91	0.77	0.73	4.2
	0.2	0.95	0.90	0.80	0.76	3.9
(256, 256)	0.5	0.95	0.92	0.70	0.78	3.9

TABLE II: Baseline comparison at $t_{win} = 0.5$ s and 0.2 s. ANN baselines are reported using multiply-accumulate operations (MACs). For the SNN, the dense input layer is reported using MACs, while subsequent spike-driven layers are reported using empirical synaptic operations (SynOps) computed from measured firing rates. Operation counts are compute proxies only and do not include memory, IO, routing, or peripheral costs.

$t_{win}(s)$	Model	F_z (N)		X, Y (mm)		Params	MACs	SynOps
		R^2	MAE	R^2	MDE			
0.5	MLP	0.91	0.79	0.69	4.7	292.9K	289.8K	–
	GRU	0.93	0.68	0.80	3.4	993.8K	14.8M	–
	LSTM	0.92	0.71	0.79	3.4	1.32M	19.8M	–
	SNN	0.92	0.73	0.82	3.3	267.8K	26.9K	1.55M
0.2	MLP	0.88	0.91	0.61	5.7	276.7K	273.7K	–
	GRU	0.89	0.83	0.66	5.0	993.8K	5.93M	–
	LSTM	0.90	0.83	0.67	4.9	1.32M	7.91M	–
	SNN	0.90	0.80	0.76	3.9	267.8K	10.8K	604.9K

Three hyperparameters are tuned via ablation: hidden layer configuration, t_{win} , and LIF membrane decay rate β .

2) *Ablation Study*: Table I reports the ablation results over hidden layer configuration, t_{win} , and β . All results are evaluated on the test set segmented using the corresponding t_{win} .

Effect of β . Reducing β from 0.95 to 0.75 at $t_{win} = 0.5$ s degrades both force and position estimation, suggesting that a higher membrane decay rate better preserves the temporal spike dynamics relevant to tactile decoding.

Effect of t_{win} . Among the tested window sizes with $\beta = 0.95$, $t_{win} = 0.5$ s yields the best position estimation, while force estimation remains consistently strong across all sizes ($R^2 \approx 0.90$ – 0.92). Shorter windows (0.2 s) likely capture insufficient temporal context, whereas longer windows (1.0 s) may incorporate transitional spike patterns from force ramping, introducing noise into the regression.

Effect of hidden layer configuration. The deeper (256, 512, 256) architecture outperforms the shallower (256, 256) network in position estimation, while the latter achieves marginally better force accuracy. This suggests that position decoding benefits from greater representational capacity, and the deeper configuration is preferred overall.

The best overall configuration (hidden layer (256, 512, 256), $t_{win} = 0.5$ s, $\beta = 0.95$) achieves an MAE of 0.73 N and MDE of 3.3 mm, both of which are acceptable for applications requiring moderate force and positional awareness. However, these ablation results reveal a discrepancy between force and position estimation. As

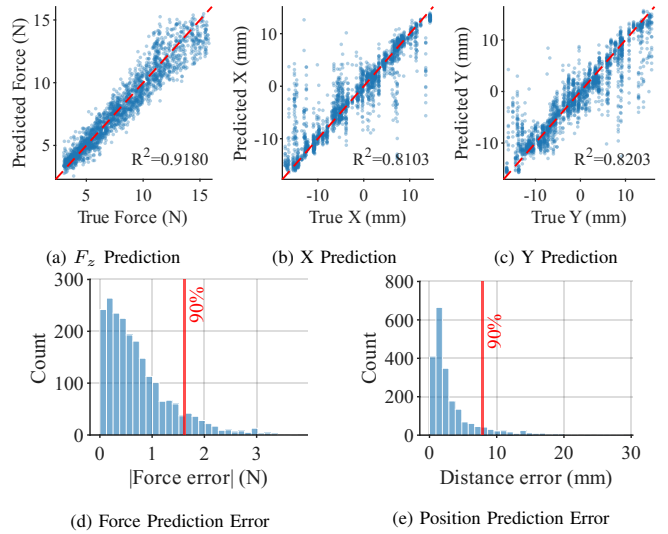


Fig. 11: Test-set regression performance of the SNN with hidden layers (256, 512, 256), $t_{win} = 0.5$ s, and $\beta = 0.95$. Predicted versus true values are shown for (a) F_z , (b) X , and (c) Y , with identity lines (red dashed) and reported R^2 . Histograms of (d) absolute force error and (e) Euclidean position error include 90th-percentile thresholds (red lines).

shown in Fig. 11, the 90th percentile force error corresponds to 10.3% of the full force range, compared to 26.3% of the maximum possible distance (30 mm) for position. Force prediction relies primarily on the overall spike firing rate, which scales monotonically with applied load and is thus relatively straightforward to regress. Position estimation, however, requires decoding both firing intensity and the spatial distribution of activity across channels — a fundamentally more complex relationship. This is evidenced by the wide spread of positional predictions at identical contact locations under varying force conditions, as shown in Figs. 11b and 11c.

3) *Comparison with Non-Spiking Baselines*: To evaluate whether the proposed SNN provides benefits beyond binned spike-count inputs, we compared it with three non-spiking baselines trained using the same sliding-window dataset, preprocessing, loss function, metrics, and train/test split described above. The MLP baseline receives the binned spike-count window flattened into a $T \times C$ feature vector, where $C = 7$ and $T = t_{win}f_{bin}$ with $f_{bin} = 30$ Hz. The flattened input is passed through fully connected layers of width 256–512–256, followed by a linear readout for $[F_z, x, y]$. The GRU and LSTM baselines instead treat the same input as a temporal sequence $\mathbf{x}_t \in \mathbb{R}^7$, $t = 1, \dots, T$, using 3-layer recurrent networks with hidden size 256 and a linear readout from the last hidden state.

As shown in Table II, models with explicit temporal structure (GRU, LSTM, and SNN) generally outperform the flattened MLP baseline, suggesting that force and position are encoded in temporal spike patterns rather than only in aggregate spike counts. Among these models, the SNN achieves comparable performance to recurrent ANN baselines at $t_{win} = 0.5$ s and the best overall performance at $t_{win} = 0.2$ s, particularly for position estimation. The SNN

also uses fewer parameters and substantially fewer dense MACs than the GRU and LSTM, with most computation represented as spike-driven SynOps. We report MACs and SynOps as architecture-level compute proxies, not system-level energy estimates, since memory access, IO, routing, and peripheral costs are hardware-dependent. These results show that the SNN preserves competitive accuracy while using a sparse event-driven computation structure compatible with neuromorphic hardware; direct energy-efficiency claims require deployment on a specific SNN accelerator.

4) *Real-time Prediction*: We evaluated the trained models on continuous spike trains to assess their suitability for real-time prediction. As shown in Fig. 12a, the model trained with $t_{\text{win}} = 0.5$ s generally tracks both force and position well, but exhibits a noticeable delay in the force response. This lag arises because the model was optimized to integrate 500 ms of spike history, effectively introducing a temporal offset proportional to the window length. For comparison, Fig. 12b shows the $t_{\text{win}} = 0.2$ s model (all other hyperparameters identical), which achieves better temporal alignment due to its shorter integration window, at the cost of increased prediction noise stemming from reduced spike accumulation.

These observations are supported quantitatively in Table III. For force evaluation, peak cross-correlation ρ_{max} is computed within a ± 1.5 s lag window to determine the optimal delay t_{lag} , followed by a lag-corrected MAE_{lag} using the aligned signals. Position error is evaluated as the per-timestep Euclidean distance between predicted and ground-truth positions, reported over the full trace (MDE_{all}) and a steady-state window (3.4–4.5 s), along with the corresponding standard deviations. Statistical outliers were excluded using a z-score criterion ($|z| \geq 2.5$). The result confirms that a narrower t_{win} minimizes force-tracking lag (as low as 2 ms for $t_{\text{win}} = 0.2$ s) and improves real-time accuracy, whereas position accuracy benefits from a moderate window size that balances the noise-suppression effect of longer temporal integration against the responsiveness penalty of predicting over an extended horizon.

D. Limitations and Future Work

This study is limited to single-point normal contact with a round indenter. Extending the sensor to shear loading or multi-touch interaction will require either additional data covering these contact modes or model-based reconstruction methods that infer pressure maps from optical signal changes, similar in concept to impedance tomography.

Although the SNN performs comparably to temporal ANN baselines such as GRUs and LSTMs, its main advantage is not fully realized with GPU-based inference. Future integration with neuromorphic accelerator hardware could process raw spike events directly, improving energy efficiency and enabling truly event-driven operation without sampling or binning.

The current manually tuned firing threshold is sensitive; automated threshold calibration would improve usability and repeatability. Future deployment studies should also evaluate

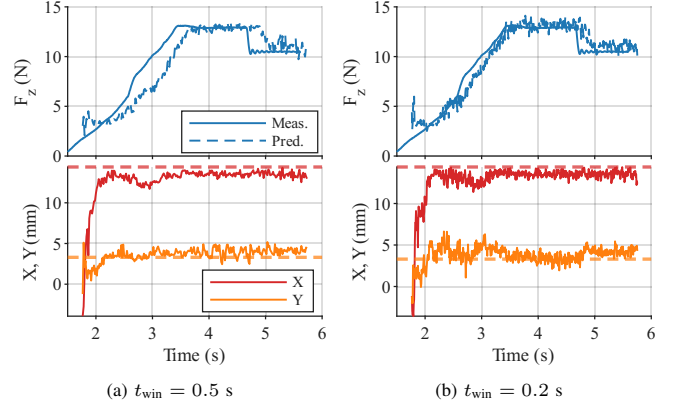


Fig. 12: Real-time force (top) and position (bottom) predictions with different training segment windows t_{win} , using hidden layers (256, 512, 256) and $\beta = 0.95$. (a) $t_{\text{win}} = 0.5$ s yields smoother but more delayed predictions, while (b) $t_{\text{win}} = 0.2$ s reduces delay at the expense of increased noise. Note that inference is performed in a streaming manner without explicit segmentation; voltage spikes are sequentially fed into the SNN model.

TABLE III: Real-time full-sequence force and localization performance across training temporal window t_{win} . Force metrics include peak cross-correlation (ρ_{max}), optimal lag (t_{lag}), and lag-corrected MAE; position metrics include mean distance error (MDE) and error standard deviation (σ) computed over the full trace and a steady-state window. Results are averaged over 107 test recordings from D10 and D12 after removing statistical outliers.

$t_{\text{win}} (s)$	$F_z (N)$			$X, Y (mm)$			
	ρ_{max}	$t_{\text{lag}} (ms)$	MAE _{lag}	MDE _{all}	MDE _{ss}	σ_{all}	σ_{ss}
0.5	0.89	34	1.13	2.60	2.13	2.24	0.91
1.0	0.75	226	1.96	3.21	2.74	1.95	1.07
0.2	0.95	2	0.81	2.99	2.49	2.65	1.11

hysteresis, long-term repeatability, temperature drift, and mechanical durability.

Further improvements may come from separately optimized force and position models, channel-wise asymmetric LED duty-cycle modulation, and optical-path redesigns that reduce spatial overlap near the sensor center to improve positional distinctiveness.

V. CONCLUSION

We presented a neuromorphic optical tactile sensor that generates event-driven spike signals directly at the analog circuit level, using only off-the-shelf IR LEDs, phototransistors, and an LIF-inspired circuit — without ADC conversion or continuous microcontroller polling. The sensor covers a 40 mm sensing area with 7 channels, encoding both contact force and location in spike firing rates and spatial activation patterns. Experimental results confirm that the sensor exhibits a clear force-dependent spiking threshold and monotonic firing rate response, while off-center contacts produce distinctive multi-channel activation signatures. A fully connected SNN trained on these spike signals achieves a force estimation mean absolute error (MAE) of 0.73 N over a 15 N range, and a contact position mean distance error (MDE) of 3.3 mm, with real-time prediction capability demonstrated across continuous pressing sequences. These

results establish that meaningful tactile information — both force magnitude and contact location — can be jointly decoded from a sparse, low-channel neuromorphic signal without requiring dense spatial coverage.

Future work will focus on hardware- and implementation-level improvements to increase spatial resolution, and on scaling the design toward larger-area robotic skin, whether as distributed patches or expanded single-sensor footprints. In parallel, we will investigate efficient communication pathways for propagating sensor-generated spikes to the central computing unit, a critical step toward realizing a full-body neuromorphic tactile system for robots.

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